

# 4DW-VLA: 4D and World-Model Aligned Vision-Language-Action for Robotic Manipulation

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**Abstract**—Most vision-language-action (VLA) policies operate on 2D images, yet robots act in a metric 3D workspace whose configuration evolves over time, a 4D problem. Methods that inject 3D or 4D structure often rely on depth sensors, point-cloud encoders, or heavy scene reconstruction, which increases model size, training cost, and deployment latency; many of them still treat spatial geometry and temporal dynamics as separate concerns. World-model-augmented VLAs predict future 2D video but rarely align that visual foresight with the robot’s own metric body motion, so predicted actions can be temporally coherent in pixel space yet physically inconsistent in 3D. We present 4DW-VLA, a 4D and world-model aligned VLA whose central component is a dedicated 4D Future Expert that predicts future 7D link trajectories (3D position + orientation) from forward kinematics (FK) at negligible cost, orders of magnitude cheaper than point-cloud or depth-pipeline 4D extraction. Unlike prior FK-based approaches that discard the 4D branch after training, the 4D Future Expert is retained at deployment and continues to supply metric body-motion context to the action head. A compact World Knowledge Expert exchanges information with the 4D expert through Unified Bidirectional Cross-Attention (UBCA), producing 2D-foresight tokens aligned with predicted 4D motion; these tokens supervise a frozen large video world model during training only. At inference, the world model is removed; the VLM, the two lightweight experts, and the action head remain, yielding a small deployable policy with no video-generation cost. On three RoboTwin 2.0 bimanual tasks, 4DW-VLA matches or exceeds representative baselines from the 2D-foresight and depth-plus-video VLA families, with the largest gains under domain randomization.

**Index Terms**—Vision-language-action models, 4D representation, world models, robotic manipulation, cross-attention.

## I. INTRODUCTION

**L**EARNING-BASED manipulation policies and VLA models [2, 3, 5, 29, 11, 1] have improved language following and cross-task generalization, but they still under-represent the physical world in which a robot moves.

**Most VLAs are fundamentally 2D.** They consume RGB images and language, yet manipulation unfolds in a 3D workspace; when time is included, the relevant object is 4D structure: how 3D geometry changes across the action horizon. A policy that reasons only in image space lacks explicit metric body motion and cannot guarantee that predicted actions respect the robot’s own spatial evolution.

**3D-/4D-aware VLAs often pay a heavy geometric tax.** Point clouds, depth maps, voxel grids, and 3D Gaussian fields can improve spatial reasoning [27, 26, 6], but they typically

require extra sensors or large perception networks to extract and encode geometry at every timestep. The resulting models are bigger, slower to train, and harder to deploy. Moreover, many 3D-enhanced designs treat geometry as a **static** per-frame cue and under-model **temporal** changes in the robot’s own configuration.

**Temporal prediction is usually misaligned across modalities.** Actions are inherently future-oriented: a policy outputs action chunks over horizon  $H$ . Yet most VLAs do not jointly predict how **2D observations** and **3D/4D body state** should co-evolve over that horizon. World-model and world-action-model (WAM) lines predict future video [7, 14, 18], but often without explicit metric 4D body motion; the camera may “imagine” a future that is not consistent with the robot’s actual 3D trajectory (e.g., the arm rises in joint space but the predicted video does not show the raised hand).

**These signals should be coupled.** When a robot lifts its end-effector in 3D, the egocentric camera stream should reflect that motion; conversely, visual foresight should not contradict feasible body dynamics. Effective manipulation therefore requires **aligning** (a) future **4D body motion** in the robot frame and (b) future **2D video** the camera will see, and using both to predict (c) future **actions**.

We propose **4DW-VLA**, a **4D and World-model aligned VLA**, to address the above gaps. Our design rests on three ideas:

- 1) **A dedicated 4D Future Expert, retained at deployment.** Instead of reconstructing scene geometry from depth or point clouds, we compute 7D link trajectories (3D position + orientation) directly from proprioception via FK, at orders-of-magnitude lower cost than perception-based 4D pipelines. Unlike prior FK-based approaches that treat 4D as a training-time auxiliary discarded at inference, the 4D Future Expert is a first-class attention path that **remains active at deployment**, continuously supplying metric body-motion structure to the action head.
- 2) **4D–2D alignment via bidirectional cross-attention.** The 4D Future Expert and a compact **World Knowledge Expert** exchange information through **Unified Bidirectional Cross-Attention (UBCA)**: at every timestep, each expert attends causally within its own stream and to past tokens of the other, coupling 4D body foresight and 2D visual foresight before action decoding.
- 3) **Train with a large frozen world model; deploy without it.** The World Knowledge Expert produces compact

**implicit 2D-foresight tokens** already aligned with 4D predictions. During training, these tokens condition a **frozen** large video diffusion world model; at inference the world model is **discarded**, so the deployable VLA stays small and fast while retaining distilled, 4D-aligned visual dynamics.

Fig. 1 summarizes the architecture. We co-train four objectives (current 3D state estimation, future 4D prediction, future 2D foresight distillation, and action-chunk generation) under one weighted multi-task loss (Section III-E). The video world model  $g_{\text{wm}}$  stays **frozen**; the VLM backbone is fine-tuned at **one-fifth** the learning rate of the expert and action heads over **10,000** optimizer steps.

#### Contributions:

- 1) We introduce a **4D Future Expert** that uses **FK-derived 7D link trajectories** as a lightweight body-centric 4D representation and is **retained at deployment**, unlike prior FK-based approaches that discard the 4D branch after training (Section III-B).
- 2) We couple the 4D Future Expert with a compact **World Knowledge Expert** through **UBCA**, aligning future metric body trajectories with future egocentric video to jointly condition chunked action prediction (Section III-C).
- 3) We distill a **frozen** video world model into trainable foresight tokens and **remove** it at inference, preserving 4D–2D alignment without deployment-time video generation (Sections III-C–III-F).

## II. RELATED WORK

### A. Foundation Models in Robotics

Generalist robot policies are trained on large multi-task datasets [25, 10, 24] and increasingly built on vision-language priors [3, 11, 1, 13]. RT-2 [3] and OpenVLA [11] cast actions as language-model tokens; Diffusion Policy [5], RDT [13], and  $\pi_0$  [1] generate continuous action chunks via diffusion or flow matching. These foundations excel at semantic generalization but largely treat observations as 2D images without explicit 4D body–video alignment.

World-action models (WAMs) co-train video prediction with control [14, 18, 16]: the predicted future regularizes the action pathway, and generation is often dropped at test time. **4DW-VLA** follows the “train with foresight, infer without the generator” principle, but differs in **what** is aligned: we explicitly couple **FK-derived 4D body trajectories** with **2D video foresight** through bidirectional cross-attention before action decoding, rather than relying on video prediction alone.

### B. 3D and 4D Representations for Manipulation

A growing line injects 3D or 4D structure into manipulation policies and VLAs. These methods improve spatial and temporal reasoning, but the **4D signals they use are typically heavy**: they require extra sensing, large encoders, auxiliary foundation models, or scene-scale reconstruction, which inflates compute, memory, and model size. **4DW-VLA** deliberately avoids this tax by deriving 4D body motion from FK and distilling 2D

dynamics into a small expert rather than embedding a full world model in the deployable policy.

**RynnWorld-4D** [28] builds a tri-branch diffusion world model that co-generates future RGB, **depth**, and **optical flow** from RGB-D input, curates hundreds of millions of frames with pseudo depth/flow labels, and trains an inverse-dynamics policy on internal 4D latents, an **infrastructure-heavy** approach. **4D-VLA** [22] injects sequential **RGB-D** frames with 3D coordinate embeddings into VLA pretraining; **depth must be available or estimated for every frame**. **Pri4R** [20] supervises VLMs with **privileged scene point tracks** from depth-based tracking, **expensive to obtain and encode**. **Lift3D-VLA** [19] lifts VLAs with explicit **point clouds**, GC-MAE pretraining over reconstructed geometry, and layer-wise temporal action modeling, adding **substantial parameters and preprocessing**. **SpatialVLA** [15] integrates **monocular depth estimation** into a 4B-scale VLA, introducing a **permanent depth branch** at inference unless distilled away. **4D-WAM** [17] aligns WAMs with dense **pixel-level 3D trajectory fields** from a large 4D foundation teacher.

In contrast, **4DW-VLA** obtains 4D through **FK on proprioception** (microseconds on CPU, no depth network, no point cloud) and aligns it to 2D foresight through **two small coupled experts**, using a frozen world model only as a **training-time teacher**. The deployable model excludes the teacher entirely.

### C. World Models and World-Action Models

Classical world models predict future states for planning or representation learning [7, 12, 8]. Recent WAMs make the future **action-facing** [14, 21]: video or latent subgoals regularize the policy [18, 16]. **4DW-VLA** contributes a complementary axis: the training signal is not only “predict future pixels” but **align future pixels with future FK 4D body motion** via bidirectional cross-attention, then condition actions on both distilled pathways. The large video model is frozen and removed at deployment, similar in spirit to fast WAM recipes but with explicit **4D–2D coupling**.

## III. METHOD

We present 4DW-VLA, a 4D and world-model aligned VLA for language-conditioned manipulation. A VLM encodes multi-view RGB and instructions; three trainable heads (a **4D Future Expert**, a **World Knowledge Expert**, and an **Action Expert**) cooperate through UBCA. Fig. 1 shows the overview.

### A. Problem Formulation

Let  $o_t$  denote multi-view RGB,  $\ell_t$  the language instruction, and  $q_t$  proprioception. Write  $p_t$  for the current metric 3D body state (FK-derived link poses),  $P_{t+1:t+H}$  for the future 4D body trajectory, and  $V_{t+1:t+T}$  for future egocentric video.

#### 4D–2D future alignment:

$$p(P_{t+1:t+H}, V_{t+1:t+T} \mid o_t, \ell_t, q_t). \quad (1)$$

#### Action prediction:

$$p(A_t \mid P_{t+1:t+H}, V_{t+1:t+T}, o_t, \ell_t, q_t), \\ A_t = [a_t, \dots, a_{t+H-1}]. \quad (2)$$

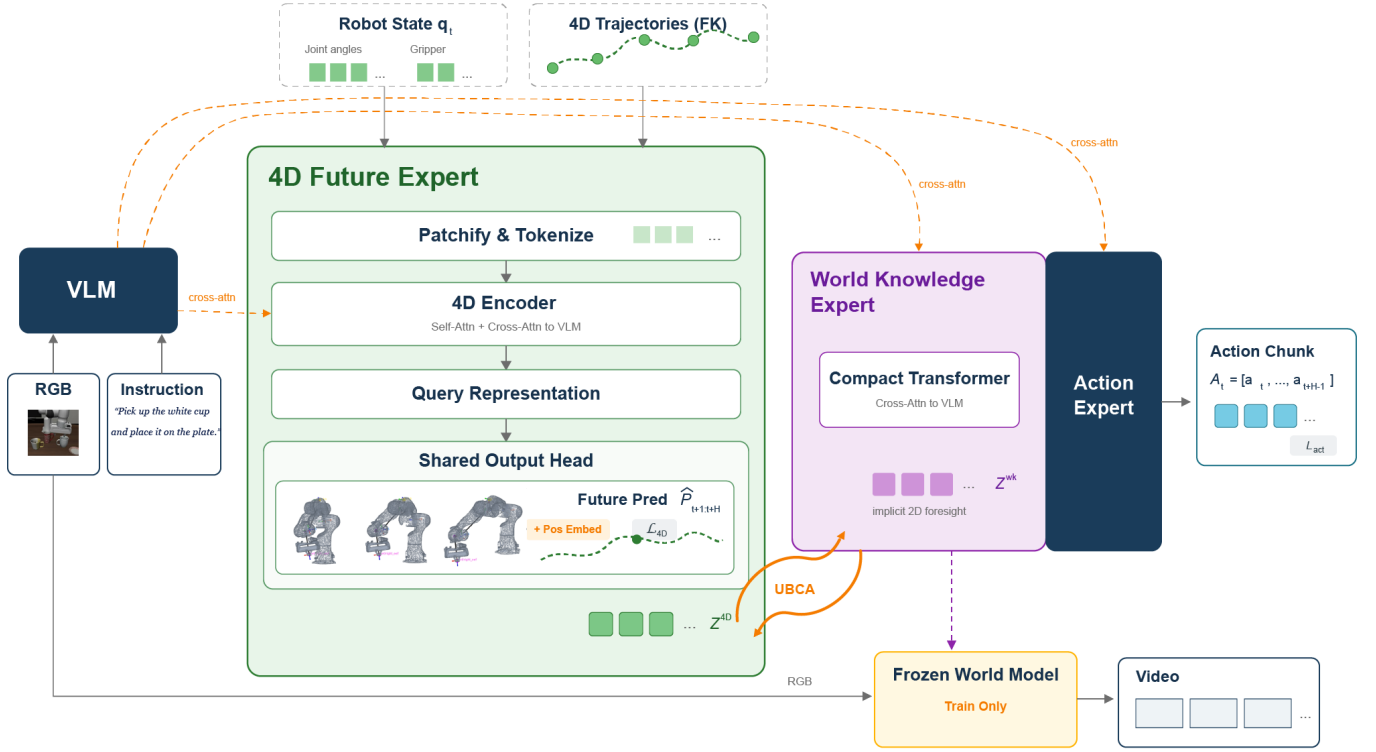


Fig. 1: **4DW-VLA architecture.** Robot state  $q_t$  and FK-derived 4D trajectories are patchified, tokenized, and fed into the **4D Future Expert**, whose 4D Encoder performs self-attention and cross-attention to the VLM prefix. A shared output head predicts both current and future 7D keypoints ( $\mathcal{L}_{4D}$ ). The resulting tokens  $Z^{4D}$  are exchanged with the **World Knowledge Expert** through UBCA, producing implicit 2D-foresight tokens  $Z^{wk}$  that supervise a **frozen** video world model during training only. The **Action Expert** attends to the VLM, both expert outputs, and generates action chunks ( $\mathcal{L}_{act}$ ). At inference the world model is dropped.

At deployment we use compact representations  $Z_t^{4D}$  and  $Z_t^{wk}$ :

$$p(A_t | Z_t^{4D}, Z_t^{wk}, o_t, \ell_t, q_t). \quad (3)$$

**Current 3D state estimation** (auxiliary):

$$p(p_t | o_t, \ell_t, P_t^{hist}), \quad (4)$$

where  $P_t^{hist}$  is a short history of FK 4D tracks.

### B. 4D Future Expert

**Motivation.** Point-cloud, depth, and scene-track 4D representations require sensing or heavy networks (Section II). Manipulators already expose  $q_t$ ; FK maps joints to link poses in microseconds. We therefore build 4D body supervision from **FK only**, gaining orders-of-magnitude lower latency and zero extra sensors compared with perception-based 4D pipelines.

**7D link trajectories.** For each link/keypoint  $k \in \{1, \dots, K\}$ ,

$$x_t^k = \text{FK}_k(q_t) = [p_t^k, r_t^k] \in \mathbb{R}^7, \quad (5)$$

where  $p_t^k \in \mathbb{R}^3$  is position and  $r_t^k \in \mathbb{R}^4$  orientation (unit quaternion) in the robot base frame.

**Expert architecture.** The 4D Future Expert is a lightweight transformer over temporal 4D tokens derived from  $q_t$  and  $P_t^{hist}$ , plus conditioning from the VLM prefix. It outputs

reconstructed **current** 4D state  $\hat{P}_t$ , predicted **future** trajectory  $\hat{P}_{t+1:t+H}$ , and hidden tokens  $Z_t^{4D}$  consumed by the World Knowledge Expert and Action Expert.

**Role of  $q_t$ .** Proprioception enters both as raw input tokens and as the quantity that defines FK targets. Conditioning on  $q_t$  anchors predictions in the robot's instantaneous configuration; through UBCA (Section III-C),  $Z_t^{4D}$  also supplies metric body context to the 2D foresight pathway.

**4D future loss:**

$$\mathcal{L}_{4D} = \beta \left( \text{MSE}(\hat{P}_t(q_t), P_t) + \gamma \text{MSE}(\hat{P}_{t+1:t+H}(q_t), P_{t+1:t+H}) \right). \quad (6)$$

**Current state loss:**

$$\mathcal{L}_{cur} = \text{MSE}(\hat{p}_t, p_t). \quad (7)$$

### C. World Knowledge Expert and 4D–2D Alignment

**Motivation.** Predicting future video alone does not guarantee consistency with body motion. We introduce a compact **World Knowledge Expert** that learns **implicit 2D-foresight tokens**  $Z_t^{wk} \in \mathbb{R}^{M \times d}$ , a low-dimensional summary of what the camera should see, **aligned** with the 4D Future Expert's predictions.

**Unified Bidirectional Cross-Attention (UBCA).** The 4D and World Knowledge experts are two transformer modules,

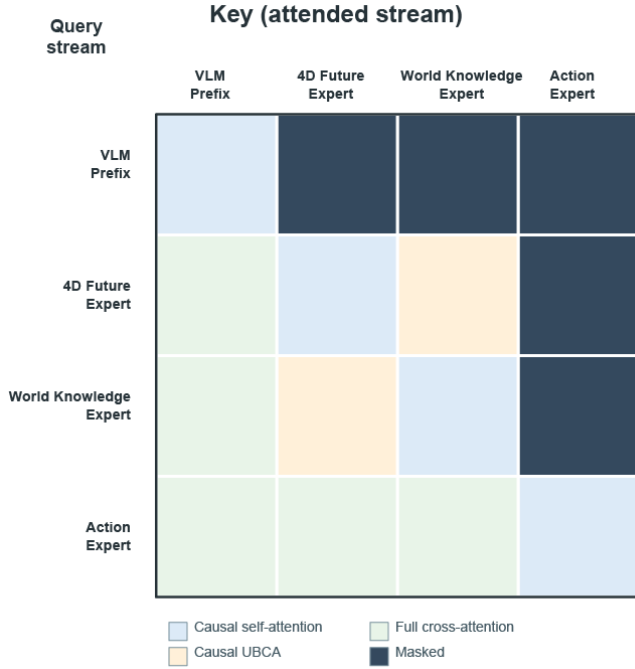


Fig. 2: **UBCA attention structure.** The mask is shown at the block level for illustration. The VLM prefix applies causal self-attention. The 4D Future Expert and World Knowledge Expert use causal self-attention within their own streams and causal cross-attention to each other, forming the bidirectional UBCA pathway. Both streams attend to the VLM prefix, while the Action Expert uses full cross-attention to all three preceding streams and causal self-attention within its own stream. Colors denote causal self-attention, full cross-attention, causal UBCA, and masked connections.

each carrying temporal tokens. At layer  $\ell$  and timestep  $i$ , for module  $A \in \{4D, wk\}$  with companion  $B$ :

$$\tilde{h}_{\ell,i}^A = \text{Attn}(q(h_{\ell,i}^A), K_{\leq i}^A, V_{\leq i}^A) + \text{Attn}(q(h_{\ell,i}^A), K_{\leq i}^B, V_{\leq i}^B), \quad (8)$$

with causal masking over timesteps: each token attends causally within its stream and to causal history in the partner stream. Fig. 2 shows the structured attention pattern.

The world-knowledge pathway produces

$$Z_t^{wk} = f_{wk}(o_t, \ell_t, q_t, Z_t^{4D}). \quad (9)$$

During training,  $Z_t^{wk}$  conditions a **frozen** video diffusion transformer  $g_{wm}$ . With VAE latents  $x_0$ , noise  $\epsilon$ , and  $\tau \sim \mathcal{U}[0, 1]$ :

$$\mathcal{L}_{vid} = \mathbb{E}_{\tau, \epsilon} [\|g_{wm}(x_\tau, h(Z_t^{wk})) - (\epsilon - x_0)\|^2]. \quad (10)$$

Only  $f_{wk}$ , UBCA parameters, and projection  $h$  are updated;  $g_{wm}$  is **frozen**.

#### D. Action Expert

The Action Expert generates action chunk  $A_t$  via flow matching, conditioned on the VLM prefix,  $Z_t^{4D}$ ,  $Z_t^{wk}$ , and

$q_t$ . With ground-truth  $A_t^0$ , noise  $\epsilon$ , and  $\tau \sim \mathcal{U}[0, 1]$ :

$$\mathcal{L}_{act} = \mathbb{E}_{\tau, \epsilon} [\|v_\theta(A_t^\tau, o_t, \ell_t, q_t, Z_t^{4D}, Z_t^{wk}) - (\epsilon - A_t^0)\|^2]. \quad (11)$$

#### E. Multi-Task Training Objective

$$\mathcal{L} = \lambda_{cur} \mathcal{L}_{cur} + \lambda_{4D} \mathcal{L}_{4D} + \lambda_{vid} \mathcal{L}_{vid} + \lambda_{act} \mathcal{L}_{act} + \lambda_{vqa} \mathcal{L}_{vqa}. \quad (12)$$

Post-training runs for **10,000** steps. The video world model  $g_{wm}$  is **frozen**. All other trainable modules (VLM, 4D Future Expert, World Knowledge Expert, UBCA, projection  $h$ , and Action Expert) are updated jointly from step 0. Let  $\eta$  denote the base learning rate for experts and the action head; the VLM uses  $\eta/5$ .

#### F. Inference

At deployment: (1) encode  $(o_t, \ell_t)$  with the VLM; (2) run both experts with UBCA to obtain  $Z_t^{4D}$ ,  $Z_t^{wk}$ ; (3) run the Action Expert to output  $A_t$ ; (4) **do not** invoke  $g_{wm}$ . The extra cost vs. a plain 2D VLA is two lightweight expert forward passes and FK on  $q_t$ .

### IV. EXPERIMENTS

We evaluate 4DW-VLA on two simulation benchmarks (RoboTwin 2.0 [4] and LIBERO-Plus) and on a real-robot platform. We ask: (1) Does the 4D Future Expert improve success over VLAs that lack explicit 3D/4D body motion? (2) Does the same recipe improve robustness under domain randomization? (3) Do gains generalize across benchmarks and transfer to real hardware?

#### A. Experimental Setup

**Benchmark.** RoboTwin 2.0 is a dual-arm manipulation suite on SAPIEN with domain randomization over clutter, lighting, background texture, table height, and language [4]. Easy = demo\_clean; Hard = demo\_randomized (OOD). Evaluations use the Aloha-AgileX bimanual embodiment; success rate is the metric.

**Tasks.** `stack_bowls_three` (geometrically precise stacking), `scan_object` (dual-arm tool use), and `place_bread_skillet` (distinct placement objective).

**Baselines.** We compare against two published VLAs that represent complementary design points; neither serves as an architectural parent of 4DW-VLA.

**InternVLA-A1.5** [9] represents the **2D-foresight VLA** family: temporal reasoning and future **2D video** distillation, but **no explicit 3D/4D body motion** or 4D–2D alignment. We re-implement and train it under the same post-training budget as 4DW-VLA.

**LingBot-VLA 2.0** [23] represents the **depth-plus-video VLA** family: **3D (depth)** cues and temporal video generation, but **no coupling** of 3D/4D body trajectories with 2D temporal evolution. We do not reproduce LingBot-VLA 2.0; numbers are **as-reported**. Not evaluated on `scan_object` or `place_bread_skillet` (n/a).

TABLE I: **Post-training success rates (%) on RoboTwin 2.0.** Easy = demo\_clean; Hard = demo\_randomized. Bold = best per column.

| Method          | stack_bowls |           | scan_obj. |           | place_bread |           |
|-----------------|-------------|-----------|-----------|-----------|-------------|-----------|
|                 | E           | H         | E         | H         | E           | H         |
| InternVLA-A1.5  | 71          | 56        | <b>45</b> | 28        | 31          | 20        |
| LingBot-VLA 2.0 | 78          | 22        | n/a       | n/a       | n/a         | n/a       |
| <b>4DW-VLA</b>  | <b>81</b>   | <b>58</b> | <b>45</b> | <b>33</b> | <b>39</b>   | <b>29</b> |

TABLE II: **Shared post-training recipe.**

| Item                       | Setting                                |
|----------------------------|----------------------------------------|
| Benchmark                  | RoboTwin 2.0                           |
| Post-training budget       | 10,000 steps                           |
| Video world model $g_{wm}$ | Frozen                                 |
| VLM learning rate          | $\eta/5$ (experts / action at $\eta$ ) |
| Easy / Hard                | demo_clean / demo_randomized           |

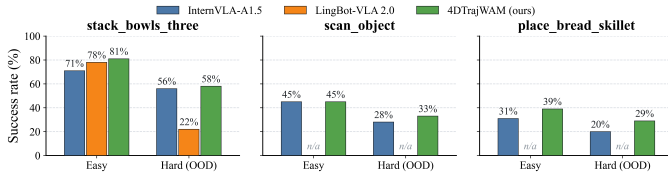


Fig. 3: **RoboTwin 2.0 success rates.** Grouped bars for Easy and Hard on three tasks. LingBot-VLA 2.0: n/a on two tasks.

**Training recipe.** 4DW-VLA and other methods share a **10,000-step** post-training budget. The video world model  $g_{wm}$  is **frozen**. Expert and action-head parameters use base learning rate  $\eta$ ; the VLM uses  $\eta/5$  (Table II).

### B. Main Results

Fig. 3 visualizes Table I.

**stack\_bowls\_three.** 4DW-VLA reaches 81%/58% (Easy/Hard) vs. 71%/56% and 78%/22% for other methods. On Easy all three are competitive; on Hard, 4DW-VLA holds 58% while one baseline collapses to 22%, consistent with **4D-aligned body foresight** under visual randomization.

**scan\_object.** 4DW-VLA ties on Easy (45%) and leads on Hard (33% vs. 28%).

**place\_bread\_skilllet.** 4DW-VLA reaches 39%/29% vs. 31%/20% for other methods (+8/+9 points). LingBot-VLA 2.0 has no reported result on this task.

4DW-VLA achieves the highest success rate in every evaluated column. Hard gains over the best baseline per task are +2, +5, and +9 points, respectively.

### C. LIBERO-Plus

To test whether the 4D Future Expert generalizes beyond RoboTwin 2.0, we evaluate on LIBERO-Plus with and without the 4D module.

### D. Real-Robot Validation

We deploy 4DW-VLA on a real bimanual platform to validate sim-to-real transfer of the FK-derived 4D signal.

TABLE III: **LIBERO-Plus success rates (%)**. Ablation of the 4D Future Expert.

| Method                | Success    |
|-----------------------|------------|
| 4DW-VLA w/o 4D Expert | TBD        |
| <b>4DW-VLA</b>        | <b>TBD</b> |

TABLE IV: **Real-robot success rates (%)**.

| Task       | 4DW-VLA w/o 4D | <b>4DW-VLA</b> |
|------------|----------------|----------------|
| TBD task 1 | TBD            | TBD            |
| TBD task 2 | TBD            | TBD            |
| TBD task 3 | TBD            | TBD            |

### E. Discussion

Easy scores match or exceed other methods on all three RoboTwin 2.0 tasks, indicating that 4D and aligned 2D auxiliary heads need not hurt in-domain learning. Consistent Hard improvements support the hypothesis that **explicit 4D-2D coupling** matters more under domain shift than **2D video foresight alone**.

## V. LIMITATIONS

- **Body-centric 4D, not full scene geometry.** FK trajectories capture the robot’s own motion, not independent object dynamics or extrinsic scene structure.
- **FK accuracy depends on calibration.** URDF-hardware mismatch propagates into 4D labels; sim-to-real kinematics gap is unstudied.
- **World-model distillation quality.** Removing  $g_{wm}$  at inference assumes  $Z_t^{wk}$  captures sufficient visual dynamics.
- **UBCA ablations pending.** Late concatenation and uni-directional cross-attention are not yet compared.
- **Limited task diversity.** Evaluation covers RoboTwin 2.0 and LIBERO-Plus; broader benchmarks and object categories remain untested.

## VI. CONCLUSION

We presented **4DW-VLA**, a **4D and world-model aligned VLA** for robotic manipulation. Its central component is a **4D Future Expert** that derives lightweight 7D body trajectories from FK and is **retained at deployment**, unlike prior FK-based approaches that discard the 4D branch after training. Coupled with a World Knowledge Expert through bidirectional cross-attention and a frozen video world model used only during training, the architecture produces a small, fast policy with 4D-aligned visual foresight and no deployment-time video generation. On RoboTwin 2.0 and LIBERO-Plus, 4DW-VLA matches or exceeds baselines that lack explicit 4D body motion, with the strongest advantage under domain randomization; real-robot experiments confirm sim-to-real transfer of the FK-derived 4D signal. Future work will ablate the 4D expert (retained vs. discarded, 7D vs. 3D) and extend to broader benchmarks.

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